

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 04: DESCRIBING MOTION & KINEMATICS

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-04

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): The displacement of an object can be zero even when the total distance travelled by it is non-zero.

Reason (R): Displacement is a vector quantity representing the shortest straight-line path between initial and final positions; returning to the start yields zero displacement.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): The magnitude of displacement can never exceed the total distance covered by a moving body.

Reason (R): Distance is the actual path length, and the straight line connecting two endpoints is always the shortest possible distance between them.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): An object moving with uniform speed can still possess acceleration.

Reason (R): Velocity is a vector quantity having both magnitude and direction; continuously changing the direction of motion causes acceleration even if speed is constant.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): The slope of a distance-time graph represents the speed of the moving object.

Reason (R): Slope is calculated as $\frac{\Delta s}{\Delta t}$, which is the rate of change of distance with respect to time.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): The area enclosed beneath a velocity-time graph and the time axis gives the displacement of the object.

Reason (R): Displacement equals velocity multiplied by time ($s = v \times t$); the product of the dimensions of the axes ($[\text{m/s}] \times [\text{s}] = [\text{m}]$) represents area.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): A car moving on a crowded city street undergoes non-uniform motion.

Reason (R): The car covers unequal distances in equal intervals of time due to traffic signals, pedestrian crossings, and turns.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.7

Assertion (A): A body can have zero velocity and non-zero acceleration simultaneously.

Reason (R): At the highest point of its vertical trajectory, an object thrown vertically upward stops momentarily ($v=0$) while gravity $g = 9.8 \text{ m/s}^2$ acts downward.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.8

Assertion (A): Negative acceleration is commonly referred to as retardation or deceleration.

Reason (R): Retardation occurs when the direction of the acceleration vector is opposite to the direction of the velocity vector, causing speed to decrease.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.9

Assertion (A): The motion of artificial satellites revolving around the Earth in circular orbits is an accelerated motion.

Reason (R): Satellites continuously change their direction of velocity towards the Earth due to centripetal gravitational acceleration.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.10

Assertion (A): Average speed can be zero for a moving body.

Reason (R): Average speed is defined as total distance travelled divided by total time taken.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): If the distance-time graph of a body is a straight line parallel to the time axis, the body is at rest.

Reason (R): The position of the body does not change with the passage of time, meaning its velocity (slope) is zero.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): An athlete running once around a circular track of radius r completes the lap in time t . His average velocity is zero.

Reason (R): The initial and final positions of the athlete coincide after one full lap, making his net displacement vector zero.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): The equations of motion ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) are valid only for motion with uniform acceleration.

Reason (R): These three kinematic equations are derived on the strict mathematical condition that acceleration a remains constant throughout.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): The velocity-time graph of a freely falling body dropped from rest is a straight line passing through the origin with a positive slope.

Reason (R): A freely falling body experiences constant downward gravitational acceleration ($g = 9.8 \text{ m/s}^2$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): An odometer installed in an automobile measures its instantaneous speed.

Reason (R): A speedometer measures instantaneous speed in km/h, whereas an odometer records total cumulative distance travelled.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): In uniform circular motion, the direction of linear velocity at any instant is along the tangent to the circular path.

Reason (R): If a stone tied to a string being whirled in a circle breaks loose, it flies off tangentially along a straight line.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): A train moving at 90 km/h is moving faster than a car moving at 25 m/s.

Reason (R): To convert km/h to m/s, multiply by $\frac{5}{18}$; $90 \times \frac{5}{18} = 25 \text{ m/s}$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): The slope of a velocity-time graph represents the acceleration of the moving body.

Reason (R): Slope is given by $\frac{\Delta v}{\Delta t}$, which is the rate of change of velocity per unit time.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): A particle travelling with constant acceleration can never reverse its direction of motion.

Reason (R): A constant force always acts in a fixed spatial direction.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): During uniform circular motion of radius R and period T , the magnitude of acceleration is $a = \frac{v^2}{R} = \frac{4\pi^2 R}{T^2}$.

Reason (R): The acceleration in uniform circular motion is directed radially inwards towards the center of curvature (centripetal acceleration).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): If the displacement-time graph of a body is a curve curving upwards with increasing slope, the body is accelerating.

Reason (R): The instantaneous slope of a position-time graph represents velocity, so an increasing slope indicates increasing velocity with time.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): A person sitting inside a train moving with uniform velocity throws an apple vertically upwards; the apple lands directly back into his hands.

Reason (R): Due to inertia of motion, the apple maintains the same horizontal forward velocity as the train and the passenger throughout its flight.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.23

Assertion (A): When a ball bounces off a hard wall with the same speed, its velocity does not change.

Reason (R): Velocity is a vector quantity that depends on both speed and direction.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.24

Assertion (A): Two stones of masses 1 kg and 5 kg dropped simultaneously from the same height in a vacuum reach the ground at the exact same instant.

Reason (R): In a vacuum, the acceleration due to gravity g is independent of the mass of the falling object ($a = g$).

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.25

Assertion (A): An object can have a northward velocity while simultaneously experiencing a southward acceleration.

Reason (R): When brakes are applied to a car heading north, the retarding acceleration vector points in the opposite direction (south).

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and correctly explains the vector definition of displacement.
Q.2	(A)	Assertion is true. Reason is true and provides the geometrical basis: $ \text{displacement} \leq \text{distance}$.
Q.3	(A)	Assertion is true. Reason is true and explains uniform circular motion where speed is constant but direction continuously turns.
Q.4	(A)	Assertion is true. Reason is true and provides the mathematical definition of slope matching speed.
Q.5	(A)	Assertion is true. Reason is true and provides the dimensional and calculus justification for area under v - t graph.
Q.6	(A)	Assertion is true. Reason is true and defines non-uniform motion in real-world scenarios.
Q.7	(A)	Assertion is true. Reason is true and provides the classic physics example of projectile motion at maximum height.
Q.8	(A)	Assertion is true. Reason is true and defines the vector relationship responsible for decelerating a moving object.
Q.9	(A)	Assertion is true. Reason is true and explains circular orbital motion as continuous centripetal acceleration.
Q.10	(D)	Assertion is false: Since distance for a moving body is always positive (>0), average speed cannot be zero (only average velocity can be zero). Reason is true.
Q.11	(A)	Assertion is true. Reason is true and explains why a horizontal slope indicates zero speed (stationary state).
Q.12	(A)	Assertion is true. Reason is true and explains why $\text{Average Velocity} = \frac{\text{Net Displacement}}{\text{Total Time}} = \frac{0}{t} = 0$.
Q.13	(A)	Assertion is true. Reason is true and highlights the vital prerequisite condition for using kinematic equations.
Q.14	(A)	Assertion is true. Reason is true and explains the linear increase in velocity $v = gt$ from rest.
Q.15	(D)	Assertion is false: The odometer records distance, not instantaneous speed (which is read by the speedometer). Reason is true.
Q.16	(A)	Assertion is true. Reason is true and provides experimental proof of tangential instantaneous velocity.
Q.17	(D)	Assertion is false: Both vehicles are moving at the exact same speed ($90 \text{ km/h} = 25 \text{ m/s}$). Reason is true.
Q.18	(A)	Assertion is true. Reason is true and provides the mathematical definition of acceleration.
Q.19	(D)	Assertion is false: An object thrown upwards has constant downward acceleration g and reverses direction at the peak. Reason is true (constant acceleration implies constant unidirectional force).
Q.20	(B)	Both statements are true, but Reason specifies the direction of centripetal acceleration rather than deriving the mathematical formula for its magnitude.
Q.21	(A)	Assertion is true. Reason is true and correctly links upward curvature on s - t graph to positive acceleration.
Q.22	(A)	Assertion is true. Reason is true and explains the relative projectile mechanics and inertia of horizontal motion.
Q.23	(D)	Assertion is false: Reversing direction completely changes the velocity vector (from $+v$ to $-v$, change $\Delta v = 2v$). Reason is true.
Q.24	(A)	Assertion is true. Reason is true and confirms Galileo's principle of universal gravitational acceleration in the absence of air resistance.
Q.25	(A)	Assertion is true. Reason is true and gives a concrete physics example where velocity and acceleration vectors point in opposite directions.